DATA SCIENCE INSTITUTE

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ASSIGNMENT

MACHINE LEARNING II

Email Campaign Classification Analysis Python

REPORT

Pillar1-e

Prepared By

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BACKGROUND:

The data is a marketing campaign data of a skin care clinic associated with its success.

Description of variables-

Success: Response to marketing campaign of Skin Care Clinic which offers both products and services. (1: email Opened, 0: email not opened)

AGE: Age Group of Customer

Recency_Service: Number of days since last service purchase

Recency_Product: Number of days since last product purchase

Bill_Service: Total bill amount for service in last 3 months

Bill_Product: Total bill amount for products in last 3 months

Gender (1: Male, 2: Female)

Note: Answer following questions using entire data and do not create test data.

QUESTIONS-

1. Import Email Campaign data. Obtain decision tree to classify cases as success=0 or 1. Obtain Sensitivity/Recall using cut-off value as 0.50 for estimated probabilities.

2. Compare performance of Decision Tree and Random Forest Method using area under the ROC curve.

3. Implement Neural Network Algorithm and obtain are under the ROC curve .

ASSIGNMENT SUMMARY OVERVIEW:

This project implements advanced machine learning algorithms including Decision Trees, Random Forests, and Neural Networks to predict email marketing campaign success for a skin care clinic. The analysis compares three distinct modeling approaches using ROC-AUC evaluation, cross-validation, and comprehensive visualization.

Key Features:

- Decision Tree with pruning and sensitivity analysis at 0.50 threshold
- Random Forest ensemble method with 100 trees
- Neural Network (MLP) with adaptive learning and early stopping
- ROC-AUC Comparison across all models
- Cross-Validation (5-Fold Stratified) for realistic performance estimates
- Feature Importance Analysis for interpretability
- Comprehensive Visualizations (10 plots including confusion matrices, ROC curves, radar charts)

Dataset: 683 customer records with demographics, purchase recency, billing history, and email response data.

Note: As per assignment requirements, models are trained on the entire dataset without train/test split. Cross-validation results are provided for more realistic performance estimates.

Seeing as we are not doing a train/test split on the data the AUC scores can be prone to being evaluated on an artifact or can be easily overfit as the evaluation would be seeing the same data from training/modelling during analysis. To circumvent this we carried out 5-fold cross validation to ensure we don't get over fitted scores that can potentially yield perfect scores of 1.0

PROJECT STRUCTURE:

📁 Project Structure

```
email-campaign-prediction/
├── data/
│   ├── raw/
│   │   └── Email Campaign.csv
│   └── processed/
│       └── email_campaign_processed.csv
├── models/
│   ├── decision_tree_model.pkl
│   ├── random_forest_model.pkl
│   ├── neural_network_model.pkl
│   ├── age_label_encoder_dt.pkl
│   ├── feature_scaler.pkl
│   └── model_metadata_dt_rf_nn.json
├── reports/
│   ├── figures/
│   │   ├── 07_decision_tree_feature_importance.png
│   │   ├── 08_decision_tree_structure.png
│   │   ├── 09_decision_tree_confusion_matrix.png
│   │   ├── 10_random_forest_feature_importance.png
│   │   ├── 11_random_forest_confusion_matrix.png
│   │   ├── 12_roc_curve_dt_vs_rf.png
│   │   ├── 13_neural_network_loss_curve.png
│   │   ├── 14_neural_network_confusion_matrix.png
│   │   ├── 15_comprehensive_model_comparison.png
│   │   └── 16_cross_validation_comparison.png
│   ├── model_comparison_dt_rf_nn.csv
│   └── cross_validation_results.csv
├── notebooks/
│   └── decision_tree_rf_nn_analysis.ipynb
├── environment/
│   ├── requirements.txt
│   └── environment.yml
├── main_tree_analysis.py
├── .gitignore
└── README.md
```

Data Pre-Processing:

Data was loaded successfully and prepared for analysis, age encoding was

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📊 Data Preprocessing
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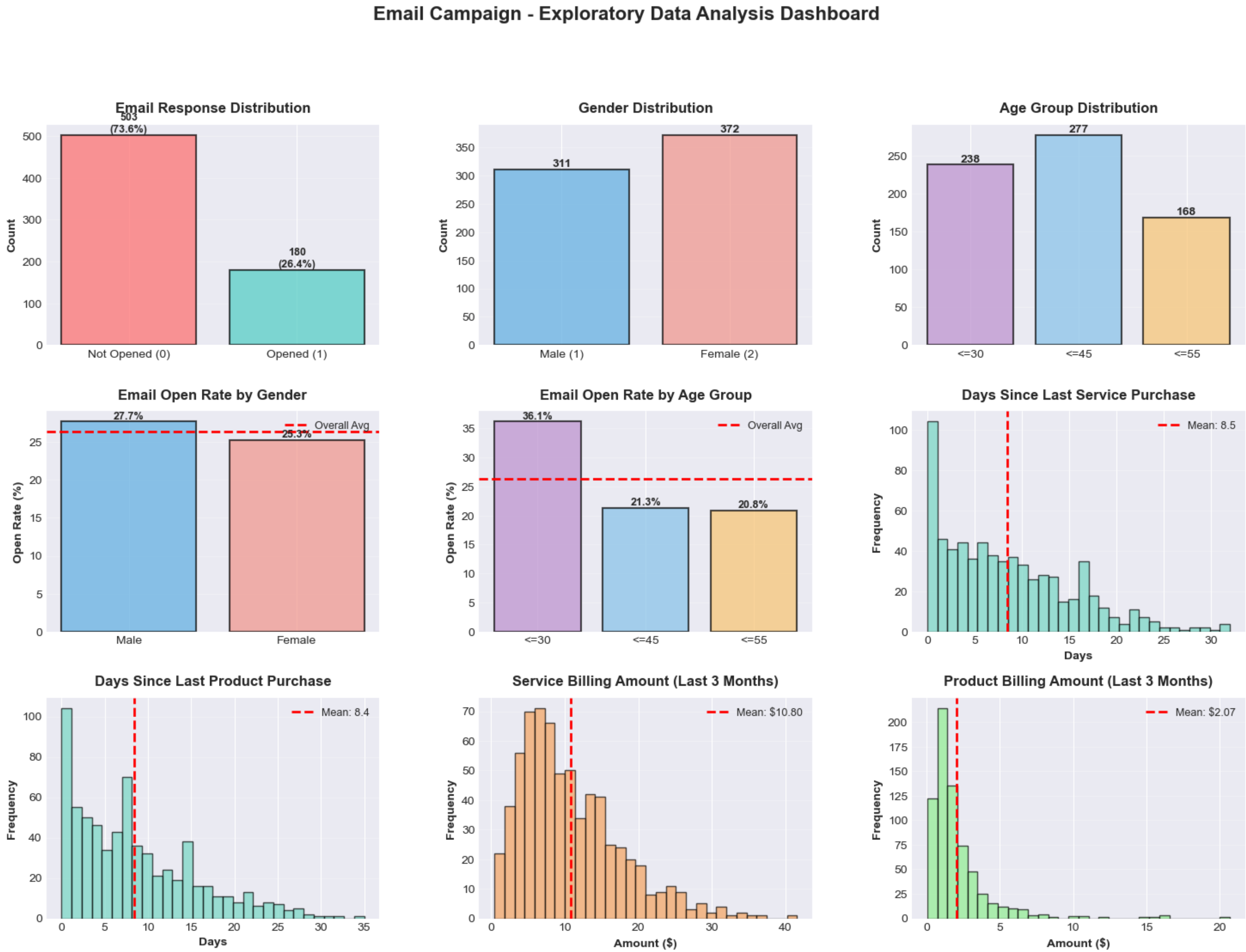
📋 Age Encoding Mapping:
   <=30 → 0
   <=45 → 1
   <=55 → 2

✅ Features prepared:
   Original features: (683, 6)
   Scaled features: (683, 6)
   Target variable: (683,)
   Class distribution: {0: 503, 1: 180}
```

EDA:



Email Campaign - Exploratory Data Analysis Dashboard
Email Response Distribution
503 (73.6%)
180 (26.4%)
Count
Not Opened (0)
Opened (1)
Gender Distribution
311
372
Male (1)
Female (2)
Age Group Distribution
238
277
168
<=30
<=45
<=55
Email Open Rate by Gender
27.7%
25.3%
Overall Avg
Open Rate (%)
Male
Female
Email Open Rate by Age Group
36.1%
21.3%
20.8%
Overall Avg
Days Since Last Service Purchase
Mean: 8.5
Frequency
Days
Days Since Last Product Purchase
Mean: 8.4
Service Billing Amount (Last 3 Months)
Mean: $10.80
Amount ($)
Product Billing Amount (Last 3 Months)
Mean: $2.07

QUESTION 1: Import Email Campaign data. Obtain decision tree to classify cases as success=0 or 1. Obtain Sensitivity/Recall using cut-off value as 0.50 for estimated probabilities.

Decision Tree Classification

Tree depth: 8

Number of leaves: 42

Number of features: 6

DECISION TREE PERFORMANCE (Cutoff = 0.50)

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📊 Classification Report:
              precision    recall  f1-score   support

  Not Opened     0.8815    0.9165    0.8986       503
      Opened     0.7375    0.6556    0.6941       180

    accuracy                         0.8477       683
   macro avg     0.8095    0.7860    0.7964       683
weighted avg     0.8435    0.8477    0.8447       683
```

SENSITIVITY/RECALL (Success=1) at cutoff 0.50: 0.6556 (65.56%)

→ Out of all actual email openers, model correctly identifies **65.56%**

Additional Metrics:

Accuracy: 0.8477 (84.77%)
Precision: 0.7375 (73.75%)
F1-Score: 0.6941

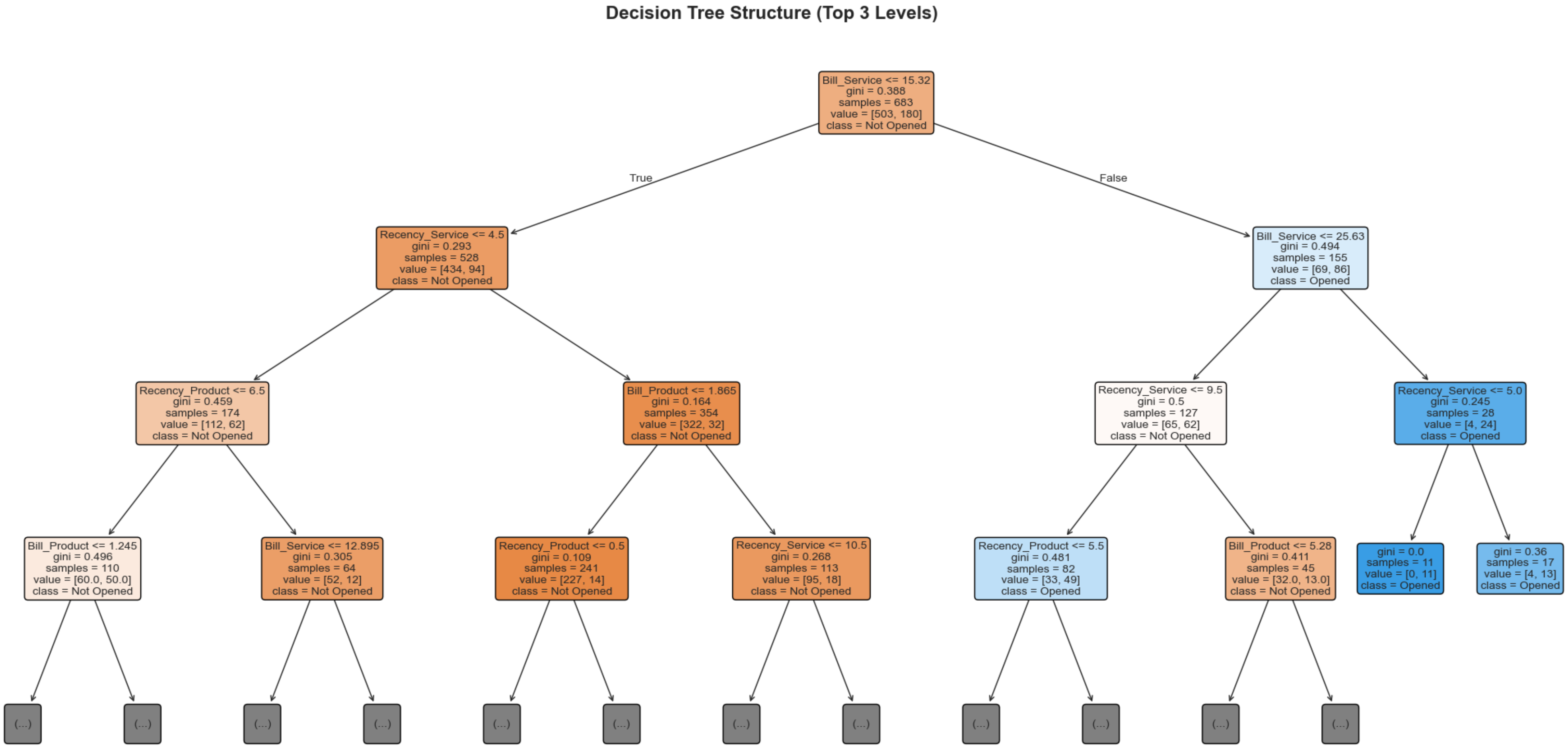
Additional Metrics:

Accuracy: 0.8477 (84.77%)
Precision: 0.7375 (73.75%)
F1-Score: 0.6941

Feature Importance:

Bill_Service:	0.400637
Recency_Service:	0.226562
Bill_Product:	0.197016
Recency_Product	0.146029
AGE_Encoded:	0.018455
Gender:	0.011301

Visualise Decision Tree



QUESTION 3: Compare performance of Decision Tree and Random Forest Method using area under the ROC curve.

Random Forest Classification

Number of trees: 100

Max tree depth: 10

Number of features: 6

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RANDOM FOREST PERFORMANCE
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📊 Classification Report:
              precision    recall  f1-score   support

  Not Opened     0.8513    0.9563    0.9007       503
      Opened     0.8136    0.5333    0.6443       180

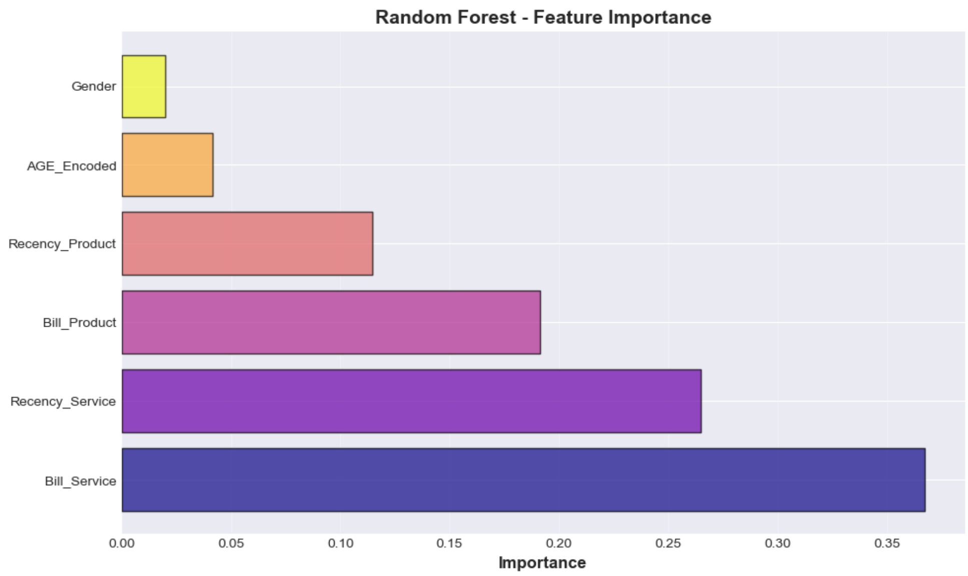
    accuracy                         0.8448       683
   macro avg     0.8324    0.7448    0.7725       683
weighted avg     0.8414    0.8448    0.8332       683
```

Performance Metrics:

Accuracy: 0.8448 (84.48%)
Precision: 0.8136 (81.36%)
Recall: 0.5333 (53.33%)
F1-Score: 0.6443

Random Forest Feature Importance:

Bill_Service: 0.367358
Recency_Service: 0.264759
Bill_Product: 0.191347
Recency_Product: 0.114864
AGE_Encoded: 0.041654
Gender: 0.020018



ROC Curve Comparison - Decision Tree vs Random Forest

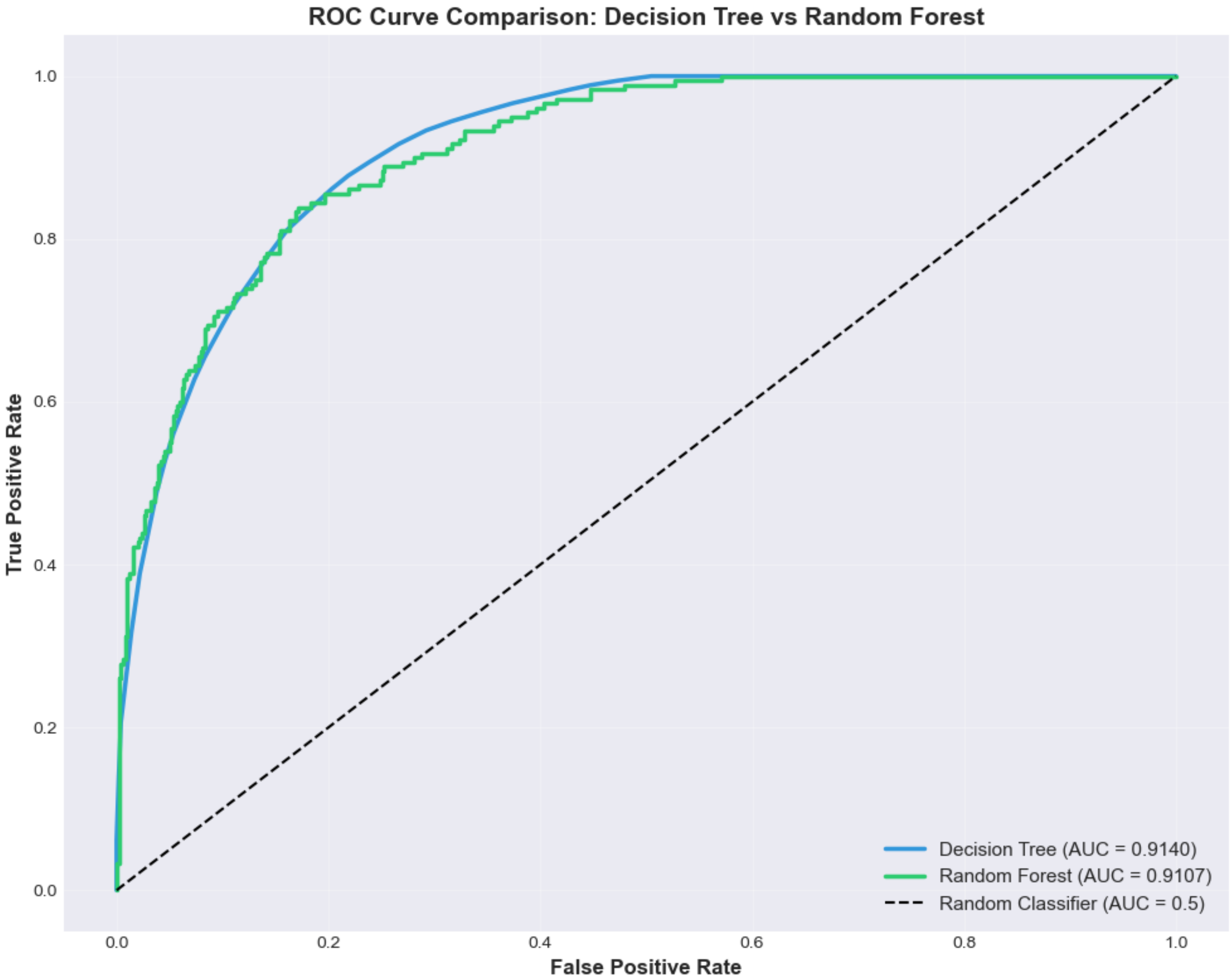
Area Under ROC Curve (AUC):

Decision Tree: 0.9140

Random Forest: 0.9107

Difference: 0.0033

Winner: Decision Tree (better by **0.33%**)



QUESTION 3: Implement Neural Network Algorithm and obtain are under the ROC curve .

Neural Network Classification

Architecture: Input → Hidden(100) → Hidden(50) → Output

Neural Network trained successfully!

Number of iterations: 12

Number of layers: 4

Final loss: 0.426709

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NEURAL NETWORK PERFORMANCE
=================================================================

📊 Classification Report:
              precision    recall  f1-score   support

  Not Opened     0.8167    0.8946    0.8539       503
      Opened     0.5985    0.4389    0.5064       180

    accuracy                         0.7745       683
   macro avg     0.7076    0.6668    0.6802       683
weighted avg     0.7592    0.7745    0.7623       683
...

📈 Area Under ROC Curve (AUC): 0.7634
```

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🏆 COMPREHENSIVE MODEL COMPARISON
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📊 PERFORMANCE SUMMARY TABLE:
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         Model  Accuracy  Precision    Recall  F1-Score       AUC  Rank
 Decision Tree  0.847731   0.737500  0.655556  0.694118  0.914032     1
 Random Forest  0.844802   0.813559  0.533333  0.644295  0.910736     2
Neural Network  0.774524   0.598485  0.438889  0.506410  0.763408     3

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🏆 BEST PERFORMING MODEL
================================================================================
   Model:      Decision Tree
   AUC:        0.9140
   Accuracy:   0.8477
   Precision:  0.7375
   Recall:     0.6556
   F1-Score:   0.6941
```

Visualisation of Comprehensive Model Comparison: